

Introduction & Network Fundamentals

Computer Network: A collection of interconnected autonomous computers. Provides resource sharing, reliability, and cost reduction.

LAN

Local Area Network - privately owned, spans a single room or building.

MAN

Metropolitan Area Network - covers a city. Example: Cable television network.

WAN

Wide Area Network - spans a large geographical area, often a country or continent.

Topologies:

- Star: All nodes connected to a central hub.
- Bus: All nodes share a single communication cable.
- Ring: Each node connects to exactly two other nodes.
- Mesh: Every node connects to every other node.

Transmission Modes:

- Simplex: Unidirectional (e.g., Keyboard to CPU).
- Half-Duplex: Both directions, but not simultaneously (e.g., Walkie-Talkie).
- Full-Duplex: Both directions simultaneously (e.g., Telephone).